

BREAST CANCER CASE STUDY OUTPUT

DATASET: SKLEARN BREAST CANCER DATASET

PS C:\Users\Aniket\DESKTOP FOLDERS\ML CASE

STUDIES\Breast_Cancer_case_study(sklearn_dataset)> &

C:/Users/Aniket/AppData/Local/Programs/Python/Python313/python.exe

"c:/Users/Aniket/DESKTOP FOLDERS/ML CASE

STUDIES/Breast_Cancer_case_study(sklearn_dataset)/BreastCancer.py"

mean radius mean texture mean perimeter ... worst concave points worst symmetry
worst fractal dimension

0	17.99	10.38	122.80 ...	0.2654	0.4601	0.11890
1	20.57	17.77	132.90 ...	0.1860	0.2750	0.08902
2	19.69	21.25	130.00 ...	0.2430	0.3613	0.08758
3	11.42	20.38	77.58 ...	0.2575	0.6638	0.17300
4	20.29	14.34	135.10 ...	0.1625	0.2364	0.07678
..	...	...		...	...	
564	21.56	22.39	142.00 ...	0.2216	0.2060	0.07115
565	20.13	28.25	131.20 ...	0.1628	0.2572	0.06637
566	16.60	28.08	108.30 ...	0.1418	0.2218	0.07820
567	20.60	29.33	140.10 ...	0.2650	0.4087	0.12400
568	7.76	24.54	47.92 ...	0.0000	0.2871	0.07039

[569 rows x 30 columns]

statistical summary of data :

	mean radius	mean texture	mean perimeter	...	worst symmetry	worst fractal
dimension	target					
count	569.000000	569.000000	569.000000	...	569.000000	569.000000
	569.000000					
mean	14.127292	19.289649	91.969033	...	0.290076	0.083946
	0.627417					
std	3.524049	4.301036	24.298981	...	0.061867	0.018061 0.483918
min	6.981000	9.710000	43.790000	...	0.156500	0.055040 0.000000
25%	11.700000	16.170000	75.170000	...	0.250400	0.071460
	0.000000					
50%	13.370000	18.840000	86.240000	...	0.282200	0.080040
	1.000000					
75%	15.780000	21.800000	104.100000	...	0.317900	0.092080
	1.000000					
max	28.110000	39.280000	188.500000	...	0.663800	0.207500
	1.000000					

[8 rows x 31 columns]

Feature Correlation on heatmap:

now feature correlation with targets

we will show with only top 10 feature :

Top 10 features correlated with target:

smoothness error	0.067016
mean fractal dimension	0.012838
texture error	0.008303
symmetry error	0.006522
fractal dimension error	-0.077972
concavity error	-0.253730
compactness error	-0.292999
worst fractal dimension	-0.323872
mean symmetry	-0.330499
mean smoothness	-0.358560

Name: target, dtype: float64

After scaling: X_scaled shape = (569, 30) y shape = (569,)

model build on training set

model build on testing set

Accuracy: 98.24561403508771

Confusion Matrix:

```
[[41  1]
 [ 1 71]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	42
1	0.99	0.99	0.99	72
accuracy		0.98		114
macro avg	0.98	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

THE END OF PROGRAM (BREAST CANCER CASE STUDY)

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STUDIES\Breast_Cancer_case_study(sklearn_dataset)>

Conclusion

- The classification model built on the Breast Cancer dataset performs **very well** with almost **perfect accuracy**.
- The results indicate that the model can be **reliably used for breast cancer tumor classification** (malignant vs benign).
- Only **2 out of 114 predictions were wrong**, showing strong generalization ability.
- With such high performance, this model can be an excellent starting point for medical diagnostic support systems, though real-world deployment should include further validation on unseen clinical data.